

COSC 1336 — Fundamentals of Programming I

Exam 2 Project Assignment - AllyBaba's Bank Transaction Manager

Project Description

Develop a program to manage **one-time bank transactions** for **AllyBaba**. The program will allow the user to input an initial account balance and then process a series of financial transactions, keeping track of deposits, withdrawals, and applicable service charges.

Functionality Overview

The program should:

1. Prompt the user to enter an **initial balance**.
2. Allow the user to enter a **series of transactions**, one at a time.
3. Each transaction must include a **transaction type**, followed by a **transaction amount** (when applicable).

Valid Transaction Types

Code	Description
------	-------------

W	Withdraw Money
---	----------------

D	Deposit Checks
---	----------------

E	End Transaction
---	-----------------

The transaction type entry should be **case-insensitive** (i.e., accept both upper and lower case).

Service Charges

- **\$0.25** service charge for each **withdrawal**.
- **\$40 overdraft charge** if a withdrawal **exceeds the current balance**.
- Withdrawals that would result in a **negative balance are not processed**.
- Service and overdraft charges are **accumulated** and **deducted at the end** of the session, **not immediately**.

Transaction Processing

After **each transaction**, display the following:

- The **transaction command and amount**
- The **resulting account balance**
- The **total service charges accrued so far**

At the **end of the session** (when the user enters E), deduct the total service/overdraft charges from the balance and display:

- **Final balance**
- **Total service charges**

Input Validation Requirements

- If the **transaction type** is invalid (i.e., not W, D, or E), display an **informative error message**, ignore the transaction, and prompt the user again.
- The **transaction amount** must be a **positive number greater than zero**. If not, display an error and prompt again.

COSC 1336 — Fundamentals of Programming I
Exam 2 Project Assignment - AllyBaba's Bank Transaction
Manager

- The program should **not allow overdraft withdrawals**—these should be rejected, and a message should be shown.

Additional Requirements

- Use **functions** to structure your program logic (e.g., `getTransaction()`, `processTransaction()`, `displaySummary()`).
- **No global variables**: all variables should be passed as parameters or returned by functions.
- All monetary values must be displayed with **2 decimal places**.

COSC 1336 — Fundamentals of Programming I
Exam 2 Project Assignment - AllyBaba's Bank Transaction
Manager

Sample Output
(User entry is in RED)

Exam 2 Project
Written by: Ally Baba

AllyBaba Transaction Statement

Today's Date: **11/2/2022**
Account Number: **123456**

Enter the beginning balance: **900**

Transaction Menu

W Process a check
D Process a deposit
E Exit

Enter transaction type: **W**
Enter Check amount: **50.00**
Processing Check for \$50.00

Processed Check for \$50.00
Current Balance: \$850.00

Service charge: \$.25 for writing a check
Accrued service charges: \$.25

Transaction Menu

W Process a check
D Process a deposit
E Exit

Enter transaction type: **W**
Enter Check amount: **1250**
Processing Check for \$1250.00

Cannot Process Check of \$1250.00
Over Draft: Insufficient Balance
Current Balance: \$850.00

COSC 1336 -- Fundamentals of Programming I
Exam 2 Project Assignment - AllyBaba's Bank Transaction
Manager

Service charge: \$40.25 for writing a check

Accrued service charges: \$40.50

Transaction Menu

W Process a check
D Process a deposit
E Exit

Enter transaction type: **D**
Enter transaction amount: **200**
Processing Deposit of \$200.00

Processed Deposit of \$200.00
Current Balance: \$1050.00

Accrued service charges: \$40.50

Transaction Menu

W Process a check
D Process a deposit
E Exit

Enter transaction type: **E**
Processing end of Transaction

Summary: End of Daily Transaction

Today's Date: 7/7/2022
Account Number: 123456

Current Balance:	\$1,050.00
Accrued service charges:	\$40.50
End of Transaction balance:	\$1,009.50

End of Exam 2 Project